John Wu

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Education

University of Maryland, College Park — College Park, MD

Bachelor of Science in Computer Science | GPA: 3.75 — *Expected Spring 2028*

- **Minors:** Robotics & Autonomous Systems; Technology Entrepreneurship & Corporate Innovation
- **Relevant Coursework:** Algorithms, Data Structures & OOP, Computer Systems (C/Assembly), Compilers, Programming Languages, Discrete Structures, Linear Algebra

Experience

Stanford University, NeuroAI Lab (Dr. Yamins) — May 2026 – Aug 2026

Research Intern — *Stanford, CA*

- Built a reusable **vision-language-action (VLA)** evaluation harness in **RoboCasa** that standardized manipulation-policy benchmarking across most LeRobot policies, now the lab's shared evaluation surface.
- Engineered a four-stage multi-camera **3D reconstruction pipeline** (calibration, FoundationStereo anchors, pose-conditioned monocular depth with spatially-varying affine correction, **4D pointcloud fusion**), recovering high-accuracy keypoints from incomplete trajectory data, then trained a multimodal model on 3D points, language, optical flow, depth, and RGB.
- Built from scratch a two-pass video captioning pipeline (full-clip semantic pass plus sliding-window atomic-motion pass), auto-labeling **100,000+ episodes** from DROID and RH20T into the lab's pretraining corpus; diagnosed and fixed a systematic lateral-motion failure with an optical-flow gripper check.
- Enabled inference on a **32B-parameter** model by sharding the checkpoint across **7 GPUs** with W8A16 quantization and PyAV decoding, and debugging a cluster GPU isolation fault to recover the full device set.

Children's National Hospital — Feb 2026 – May 2026

ML Engineering Intern — *Washington, DC*

- Delivered the core computer-vision pipeline for a mobile app that scores youth-football tackle severity from game film, giving clinicians an objective tool for early injury screening.
- Converted raw game film into normalized per-joint kinematics and automatic impact-frame detection by architecting an end-to-end pipeline of **YOLOv8** + **ByteTrack** tracking, **YOLOv8-pose**, and **VideoPose3D** lifting.
- Improved detection reliability over off-the-shelf weights by assembling and augmenting a custom 1,000+ clip dataset and running supervised fine-tuning; open-sourced at github.com/JohnW-07/tackle-vision-ML.

Leuterio Thomas — Jul 2025 – Aug 2025

Software Engineering Intern — *National Harbor, MD*

- Cut physical model production from ∼3 hours to ∼15 minutes by building **C#** plugins on the **Revit API** that converted client **BIM** models into print-ready geometry, stripping extraneous detail, color-coding elements by discipline, and slicing to printable layers.
- Generalized the pipeline across differing model structures rather than hand-tuning per job, and ran it to produce demonstration models for 5–6 client projects.

Projects

Terps Racing EV @ UMD — Sep 2025 – Dec 2025

Lead Engineer, Vehicle Dashboard — *College Park, MD*

- Delivered live vehicle telemetry at 60 Hz across dozens of **CAN** sensors by leading 6 engineers to build a CAN-bus data-acquisition platform and driver interface.
- Cut boot time **75%** (from ∼20s to under 5s) by trimming dependencies and deferring loading on a crash-resilient, RAM-only **piCore** OS.
- Kept the build on schedule by delegating subsystems, driving technical pivots, and teaching the team the embedded stack.

ArmToPoint (ConvPose) — Sep 2025 – Present

ML Engineer

- Predicted 3D object position to within ±**5 cm** from a single RGB camera by training a lightweight **PyTorch** CNN on **MuJoCo**-rendered images.
- Enabled on-device use (full videos processed in under 1 minute on a phone) by optimizing for edge inference and feeding predictions into a control loop driving a simulated arm's end-effector.

Cherry Canary — Bitcamp 2026

Full-Stack & Hardware | ***1st Place, Social Good Track*** — *College Park, MD*

- Shipped a wearable respiratory monitor end to end in 36 hours, owning **ESP32** firmware and streaming live sensor data to a **React** dashboard through a **FastAPI** backend on **MongoDB Atlas**.

Technical Skills

Languages: Java, Python, C, C++, C#, TypeScript/JavaScript, SQL
Backend & Data: FastAPI, Node.js, REST APIs, MongoDB, SQLite, Git, Linux
ML & Computer Vision: PyTorch, NumPy, OpenCV, YOLO, ByteTrack, VideoPose3D, vision–language models (VLM/VLA)
Robotics & Simulation: MuJoCo, RoboCasa, CAN bus, ESP32, RevitAPI, Unity3D, multi-GPU clusters